MIT 18.06SC - 1.8

Problem 7.1

a) $A = \begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 2 & -2 & 11 & -3 \end{bmatrix}$

$$\downarrow$$

$$\begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 0 & -12 & -3 & -21 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 1 & 1/4 & 7/4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\downarrow$$

$U = \begin{bmatrix} 1 & 0 & 23/4 & 1/4 \\ 0 & 1 & 1/4 & 7/4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

b) Rank = 2 (as there are two pivot columns)

[redacted]

c) The nullspace matrix N is:-

$$\begin{bmatrix} -23/4 & -1/4 \\ -1/4 & -7/4 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The special solutions are its columns i.e. $\begin{bmatrix} -23/4 \\ -1/4 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1/4 \\ -7/4 \\ 0 \\ 1 \end{bmatrix}$

Problem 7.2

Let $A_1 = \begin{bmatrix} 5 & 4 \\ 10 & 8 \end{bmatrix}$, then $A_1B = \begin{bmatrix} 9 & 9 \\ 18 & 18 \end{bmatrix}$ and so rank $(A_1B) = 1$ because row 2 = 2 x row 1 and so there is only one pivot colum.

Let $A_2 = \begin{bmatrix} 5 & -5 \\ 10 & -10 \end{bmatrix}$, then $A_2B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ and so rank $(A_2B) = 0$ since A_2B is the ~~zero~~ 0 matrix.